## Tal Boger

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### Education

Ph.D., Psychological & Brain Sciences, Johns Hopkins University — 2023 — present
Advisor: Chaz Firestone

BA, Statistics & Data Science, Yale University — 2019 — 2023
*with distinction*

### Publications (* equal contribution)

*Journal Articles*

**Boger, T.** (in press). Compression without compromise [commentary]. *Behavioral and Brain Sciences*.

**Boger, T.**, & Firestone, C. (in press). Complexity is a unified cognitive kind. *Nature Human Behaviour*.

**Boger, T.**, & Firestone, C. (2026). Perceiving animacy in 'identical' images. *eLife*, *15*, RP111179.

**Boger, T.**, & Firestone, C. (2025). The psychophysics of style. *Nature Human Behaviour*, *9*(12), 2497-2509.

**Boger, T.**, & Firestone, C. (2025). Visual anagrams reveal high-level effects with 'identical' stimuli. *Current Biology*, *35*(19), PR910-PR912.

***Boger, T.**, *Yousif, S. R., McDougle, S. D., & Rutledge, R. B. (2025). Random behavior is stable across tasks and time. *Journal of Experimental Psychology: General, 154*(6), 1571-1582.

**Boger, T.**, & Strickland, B. (2025). Object persistence explains event completion. *Cognition*, *259*, 106110.

**Boger, T.**, & Keil, F. C. (2025). Mechanistic complexity is fundamental: Evidence from judgments, attention, and memory. *Journal of Experimental Psychology: General*, *154*(2), 596-606.

**Boger, T.**, & Franconeri, S. L. (2024). Reading a graph is like reading a paragraph. *Journal of Experimental Psychology: General*, *153*(7), 1699-1704.

Li, Y., Wang, Y.Q., **Boger, T.**, Smith, K., Gershman, S.J., & Ullman, T. (2023). An approximate representation of objects underlies physical reasoning. *Journal of Experimental Psychology: General*, *152*(11), 3074-3086.

**Boger, T.**, & Ullman, T. (2023). What is "where": Physical reasoning informs object location. *Open Mind, 7*, 130-140.

*Hafri, A., ***Boger, T.**, & Firestone, C. (2022). Melting ice with your mind: Representational momentum for physical states. *Psychological Science, 33*(5), 725-735.

*Refereed Conference Proceedings*

**Boger, T.**, Most, S.B., & Franconeri, S.L. (2021). Jurassic mark: Inattentional blindness for a datasaurus reveals that visualizations are explored, not seen. In *2021 IEEE Visualization Conference (VIS)* (pp. 71-75). (**Best Short Paper Award**.)

**Boger, T.**, Ananthabhotla, I., & Paradiso, J. (2021). Manipulating causal uncertainty in sound objects. In *Proceedings of the 16th International Audio Mostly Conference* (pp. 9-15).

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**Conference Presentations**

**Boger, T.**, & Firestone, C. (2026). Perceiving animacy in 'identical' images. Poster presented at the annual meeting of the *Society for Philosophy and Psychology*, 6/18/2026, Baltimore, MD.

**Boger, T.**, & Firestone, C. (2026). Perceiving animacy in otherwise-identical images. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/18/2026, St. Pete Beach, FL.

Firestone, C., & **Boger, T.** (2026). Phrasal momentum. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/17/2026, St. Pete Beach, FL.

**Boger, T.**, Firestone, C., & Liu, S. (2026). Who drew this? The development of visual style perception. Poster presented at the biennial meeting of the *Cognitive Development Society*, 4/10/2026, Montreal, Canada.

**Boger, T.**, Firestone, C., & Liu, S. (2025). Who drew this? Children appreciate visual style differently than adults. Poster presented at the annual meeting of the *Cognitive Science Society*, 8/2/2025, San Francisco, CA.

**Boger, T.**, & Firestone, C. (2025). Complexity is a unified cognitive kind. Talk given at the annual meeting of the *Society for Philosophy and Psychology*, 6/19/2025, Ithaca, NY.

Liu, S., Firestone, C., & **Boger, T.** (2025). Who drew this? Children appreciate visual style differently than adults. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/20/2025, St. Pete Beach, FL.

**Boger, T.**, Liu, S., & Firestone, C. (2025). Complexity is a cognitive universal: Evidence from cross-modal transfer. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/17/2025, St. Pete Beach, FL.

Firestone, C., & **Boger, T.** (2025). Controlling for everything: Canonical size effects with identical stimuli. Talk given at the annual meeting of the *Vision Sciences Society*, 5/17/2025, St. Pete Beach, FL.

**Boger, T.**, & Firestone, C. (2024). The psychophysics of style. Talk given at the annual meeting of the *Object Perception, Attention, and Memory Conference*, 11/21/2024, New York, NY. (**Best Talk Award**.)

**Boger, T.**, & Firestone, C. (2024). The psychophysics of style. Talk given at the annual meeting of the *Society for Philosophy and Psychology*, 6/20/2024, West Lafayette, IN. (**William James Prize**.)

**Boger, T.**, & Firestone, C. (2024). The psychophysics of style. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/18/2024, St. Pete Beach, FL.

Franconeri, S. L., & **Boger, T.** (2024). When processing relationships, visual processing capacity is far less than four. Talk given at the annual meeting of the *Vision Sciences Society*, 5/18/2024, St. Pete Beach, FL.

Hafri, A., **Boger, T.**, & Firestone, C. (2022). Remembering the future: Representational momentum for physical states. Talk given at the annual meeting of the *(European) Society for Philosophy and Psychology*, 7/21/2022, Milan, Italy.
**Boger, T.**, & Keil, F. C. (2022). Mechanism influences intuitive object complexity. Talk given at the annual meeting of the *(European) Society for Philosophy and Psychology*, 7/19/2022, Milan, Italy.

**Boger, T.**, & Firestone, C. (2022). Automatic simulation of unseen physical events. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/17/2022, St. Pete Beach, FL.

**Boger, T.**, Most, S.B., & Franconeri, S.L. (2021). Jurassic mark: Inattentional blindness for a datasaurus reveals that visualizations are explored, not seen. Talk given at the annual (online) meeting of the *IEEE Visualization Conference (VIS)*, 10/26/2021.

**Boger, T.**, Ananthabhotla, I., & Paradiso, J. (2021). Manipulating causal uncertainty in sound objects. Talk given at the annual (online) meeting of the *Audio Mostly Conference*, 9/1/2021.

**Boger, T.**, & Firestone, C. (2021). Through the looking-glass: Visual sensitivity to chirality. Poster presented at the annual (online) meeting of the *Vision Sciences Society*, 5/25/2021.

Hafri, A., **Boger, T.**, & Firestone, C. (2021). Melting ice with your mind: Dynamic representation of physical states. Talk given at the annual (online) meeting of the *Vision Sciences Society*, 5/25/2021.

**Boger, T.**, Hafri, A., & Firestone, C. (2020). Melting ice with your mind: Representational momentum in state-space. Talk given at the annual (online) meeting of the *Object Perception, Attention, and Memory Conference*, 11/19/2020.

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**Honors & Awards**

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| 2026 | Walter L. Clark Teaching Award, *Johns Hopkins Psychological & Brain Sciences* |
| 2025 | Robert S. Waldrop Jr. Investigator Award, *Johns Hopkins Psychological & Brain Sciences* |
| 2024 | Best Talk Award, *Object Perception, Attention, and Memory Conference* |
| 2024 | Professional Development Award, *Object Perception, Attention, and Memory Conference* |
| 2024 | William James Prize, *Society for Philosophy and Psychology* |
| 2023 | Graduate Research Fellowship Program (GRFP), *National Science Foundation* |
| 2022 | National Eye Institute Early Career Scientist Travel Grant, *Vision Sciences Society* |
| 2021 | Best Short Paper Award, *IEEE Visualization Conference (VIS)* |

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**Invited Talks**

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| 2025 | Cognitive Tools Lab (Fan lab), *Stanford University* |
| 2024 | Developing Minds Lab (Kibbe lab), *Boston University* |
| 2023 | Computational Cognitive Neuroscience Lab (Gershman lab), *Harvard University* |
| 2023 | Computation and Language Lab (Piantadosi lab), *UC Berkeley* |
| 2022 | Vision Sciences Lab (Konkle and Alvarez lab), *Harvard University* |
| 2021 | Machine Common Sense Meeting, *DARPA* |

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**Teaching Experience**

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| *Instructor* | The art of information: Information-theoretic approaches to psychology and art (Intersession, 2026) |
| *Teaching Assistant* | Introduction to Psychology (Fall, 2026)<br>Introduction to Psychology (Fall, 2025)<br>Advanced Research Design and Analysis (Spring, 2025)<br>Advanced Statistical Methods (Fall, 2024) |
| *Guest Lecturer* | Introduction to Psychology (Fall, 2025; Fall, 2026)<br>Advanced Statistical Methods (Fall, 2024) |

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**Professional Service and Membership**

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| *Ad-Hoc Reviewer* | Journals: Attention, Perception, & Psychophysics; Cognition; Journal of Experimental Psychology: Human Perception and Performance; Open Mind; Perception; Psychonomic Bulletin & Review; IEEE Transactions on Visualization and Computer Graphics<br><br>Conferences: Cognitive Science Society; ACM Conference on Human Factors in Computing Systems (SIGCHI) |
| *Societies* | Cognitive Development Society; Cognitive Science Society; Society for Philosophy and Psychology; Vision Sciences Society |